

EXP 1

Aim

To install and configure Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium and verify that the required environment is working correctly for Reinforcement Learning experiments.

Algorithm

1. Start the installation process.
2. Download and install Anaconda Navigator on the system.
3. Open Anaconda Navigator and create a new Python environment.
4. Select the required Python version for the environment.
5. Activate the environment using Anaconda Prompt.
6. Install TensorFlow using the appropriate package installation command.
7. Install Keras and its required dependencies.
8. Install Gymnasium for Reinforcement Learning environments.
9. Launch Jupyter Notebook from the created Anaconda environment.
10. Import and verify TensorFlow, Keras, and Gymnasium.
11. Display their versions to confirm successful installation.
12. Stop the process after successful verification.

CODE

```
import sys

import numpy as np

import tensorflow as tf

import keras

import gymnasium as gym

import matplotlib.pyplot as plt

print("=" * 60)

print("  REINFORCEMENT LEARNING SOFTWARE CONFIGURATION")

print("=" * 60)

python_version = sys.version.split()[0]

numpy_version = np.__version__

tensorflow_version = tf.__version__

keras_version = keras.__version__

gym_version = gym.__version__

print("Python Version   :", python_version)

print("NumPy Version    :", numpy_version)

print("TensorFlow Version :", tensorflow_version)
```

```
print("Keras Version :", keras_version)
print("Gymnasium Version :", gym_version)
print("\n" + "=" * 60)
print("      INSTALLATION STATUS")
print("=" * 60)
libraries = [
    "Python",
    "NumPy",
    "TensorFlow",
```

```

    "Keras",
    "Gymnasium"
]
status = [1, 1, 1, 1, 1] for
library in libraries:
    print(library, ": Successfully Configured")
print("\nAll required libraries are installed successfully!")
plt.figure(figsize=(9, 5))
plt.bar(libraries, status)
plt.xlabel("Software / Library")
plt.ylabel("Configuration Status")
plt.title("Reinforcement Learning Software Configuration") plt.ylim(0,
1.2)
plt.yticks([0, 1], ["Not Configured", "Configured"])
plt.xticks(rotation=20)
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
print("\n" + "=" * 60)
print("
                RESULT"
) print("=" * 60)
print("Python, NumPy, TensorFlow, Keras and Gymnasium")
print("were successfully configured for Reinforcement Learning.") print("\nExperiment 1
Completed Successfully!")

```

OUTPUT

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REINFORCEMENT LEARNING SOFTWARE CONFIGURATION

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Python Version : 3.12.13

NumPy Version : 2.0.2

TensorFlow Version : 2.20.0

Keras Version : 3.13.2

Gymnasium Version : 1.3.0

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INSTALLATION STATUS

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Python : Successfully Configured

NumPy : Successfully Configured

TensorFlow : Successfully Configured

Keras : Successfully Configured

Gymnasium : Successfully Configured

All required libraries are installed successfully!